

ML in Applications

***Dipartimento di Automatica e Informatica
Politecnico di Torino, Torino, ITALY***



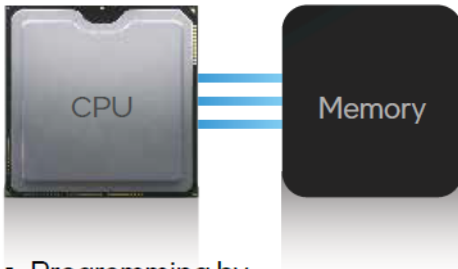
**Politecnico
di Torino**

Bioinformatics & Neuromorphic Lab



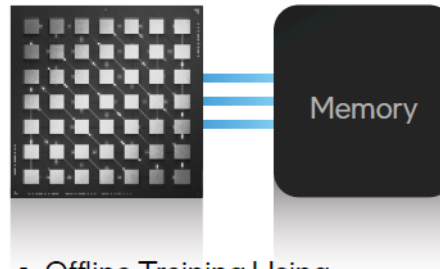
Today's Computing Architectures

Conventional Computing



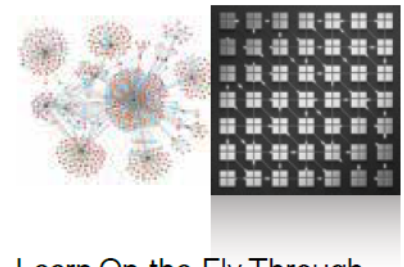
- Programming by Encoding Algorithms
- Synchronous Clocking
- Sequential Threads of Control

Parallel Computing



- Offline Training Using Labeled Datasets
- Synchronous Clocking
- Parallel Dense Compute

Neuromorphic Computing



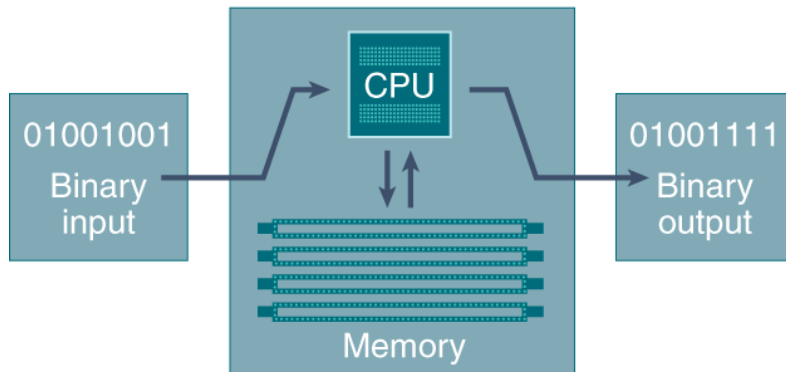
- Learn On-the-Fly Through Neuron Firing Rules
- Asynchronous Event-Based Spikes
- Parallel Sparse Compute

adapted from:

[Taking Neuromorphic Computing with Loihi 2 to the Next Level Technology Brief \(intel.com\)](https://www.intel.com/content/www/us/en/ai/neuromorphic/taking-neuromorphic-computing-with-loihi-2-to-the-next-level-technology-brief.html)

Digital vs Neuromorphic Computing

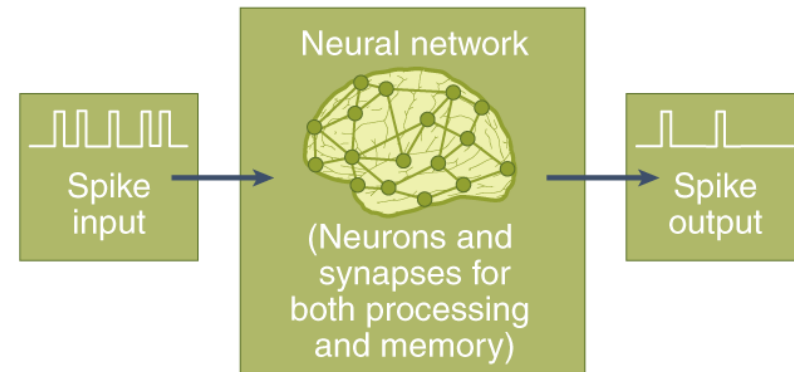
Von Neumann architecture



Sequential processing
Separated computation and memory
Code as binary instructions
Binary data
Synchronous (clock-driven)

versus

Neuromorphic architecture



Massively parallel processing
Collocated processing and memory
Spiking neural network
Spikes
Asynchronous (event-driven)

Operation
Organization
Programming
Communication
Timing

adapted from:
[Schuman et al. Nature 2022](#)

Neuromorphic Computing vs Digital Computing

Asynchronous sparse event driven computing

Neuromorphic Architecture

Neuromorphic Computing

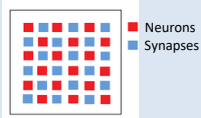
Animal Brain

Configuration/Logic

Bio-inspired local Learning Algorithms

Unknown

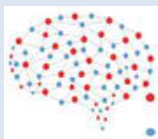
Algorithm



Distributed Processing & memory



Sparse & Asynchronous Neural Networks



Brain computing system

Architecture

Address Event Representation and/or Analog Signals

Spiking Signals (space & time)

Spiking Signals

Encoding Scheme

Analog/digital processing elements

Bio-inspired Artificial Neurons and synapses

Biological Neurons & synapses

Devices

Neuromorphic Sensors (Silicon Retina, Silicon Cochlea)

Biological Sensors (Eyes, ears, tactile)

Sensing

HW

SW

HW

SW

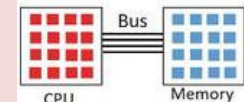
Synchronous computing

Digital Computer Architecture

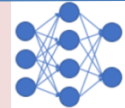
Deep Learning

Program/Logic

Global Learning Algorithms



von Neumann Architecture



Neural Networks Fully connected & Synchronous

Binary Signals

Real Value (space or time)

CPU (Logic Gates) Memory (SRAM)

Artificial Neurons Math functions (Relu, sigmoid)

HW

SW

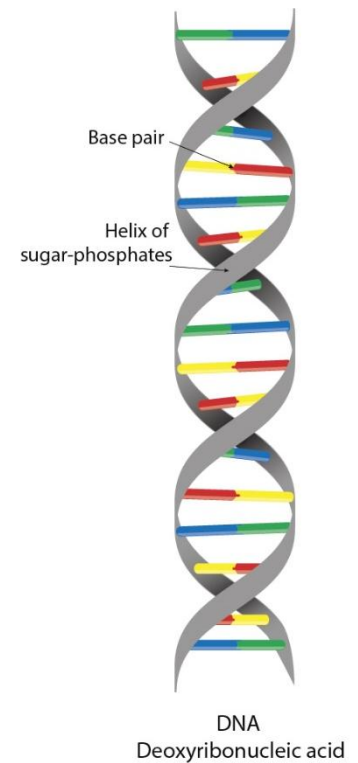
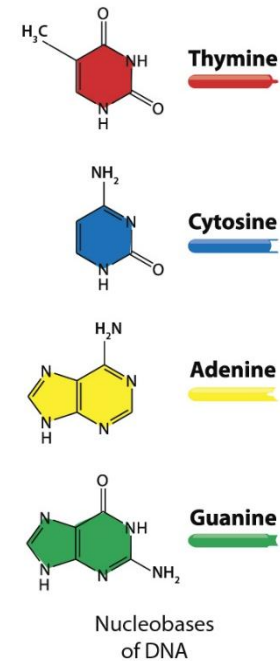
ML in applications

Neuromorphic Computing & Bioinspired Spatial data

Bioinformatics

The language of DNA is digital, but not binary

- DNA has **4 positions**, T, C, G and A
- DNA '**byte**' (called a '**codon**') has **three digits**
- DNA codon has **64 possible values**

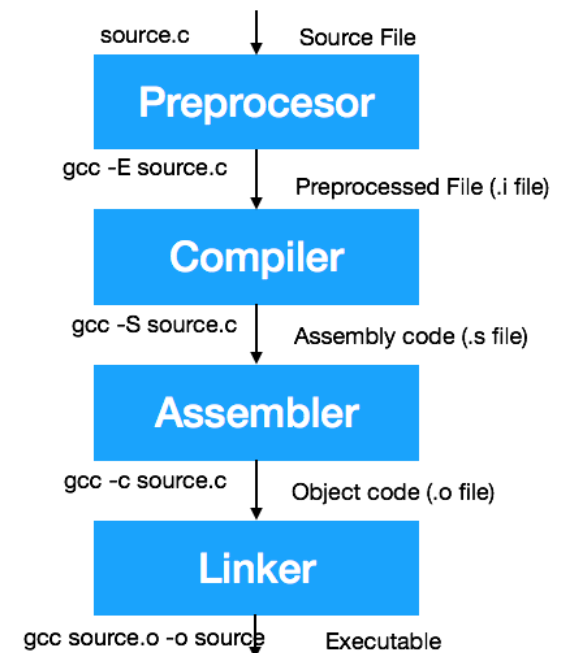
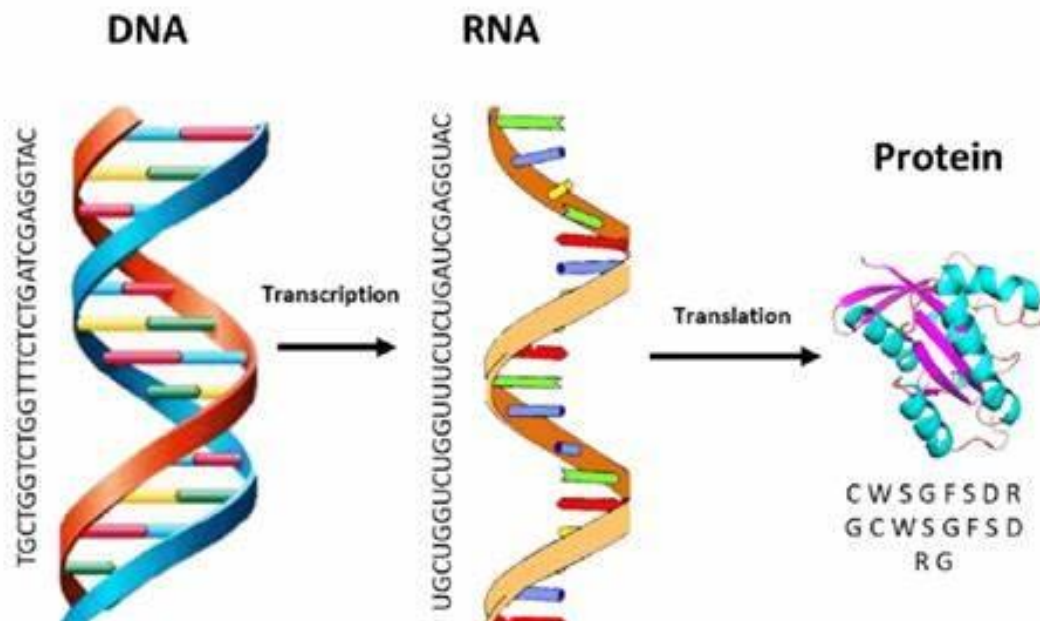


Bioinformatics

The Central Dogma compared with compiled programs

DNA is used to make **RNA** that is used to make **proteins**

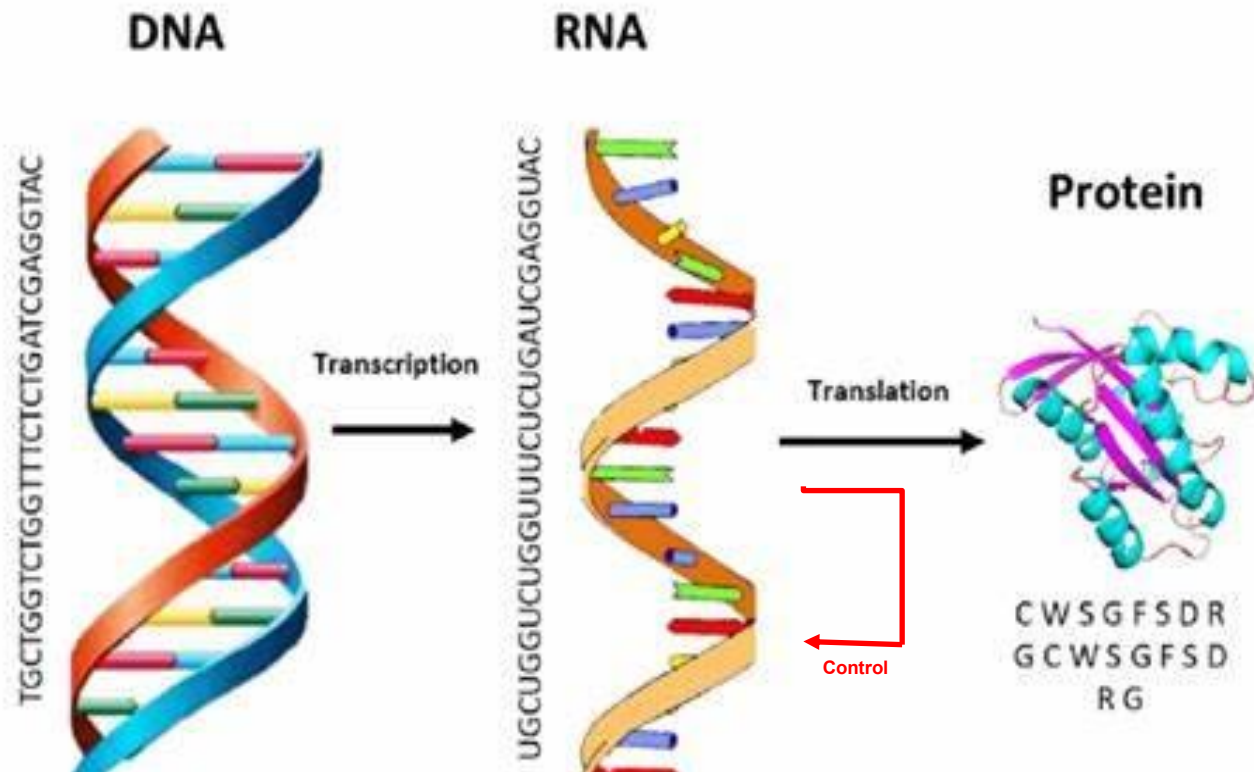
From a **.c file** comes a **.o** object file, which can be compiled into an **executable** (a.out/exe)



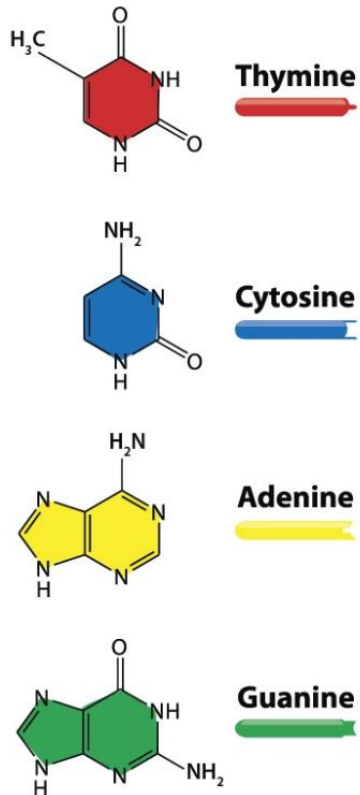
Bioinformatics

The Central Dogma V2.0

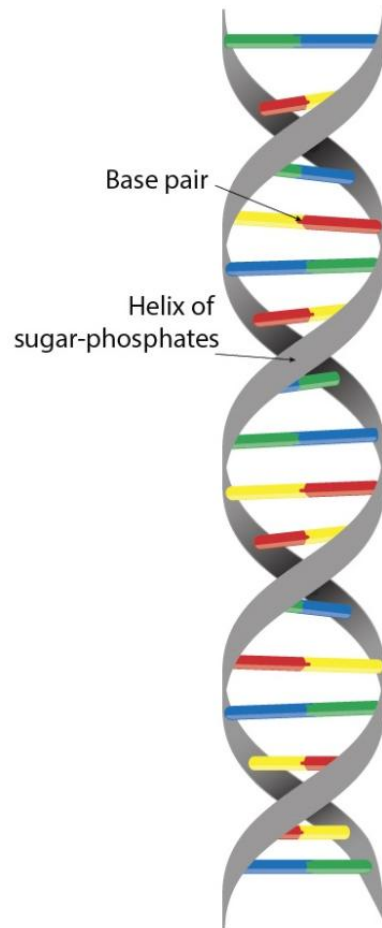
DNA is used to make **RNA** that is used to make **proteins** **and to control the protein translation**



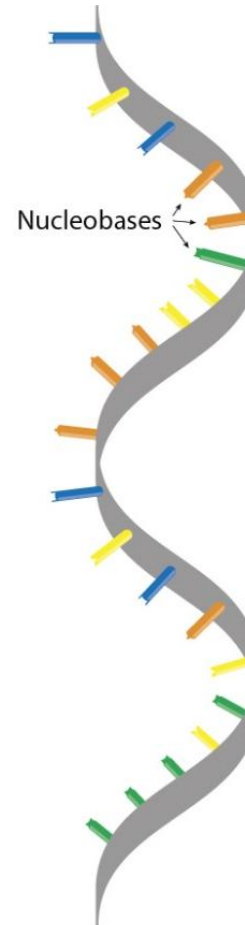
Bioinformatics



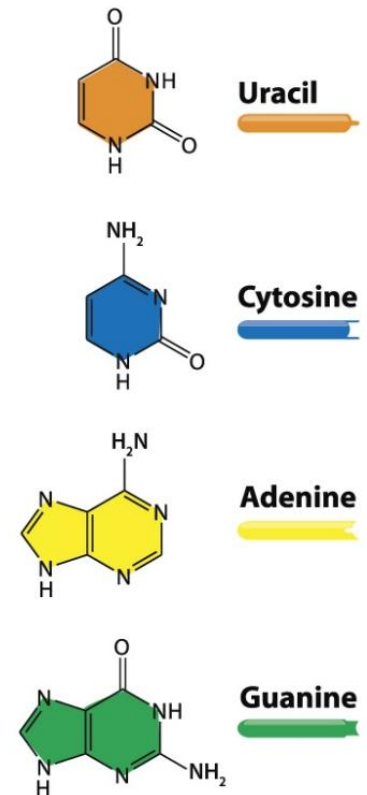
Nucleobases
of DNA



DNA
Deoxyribonucleic acid



RNA
Ribonucleic Acid

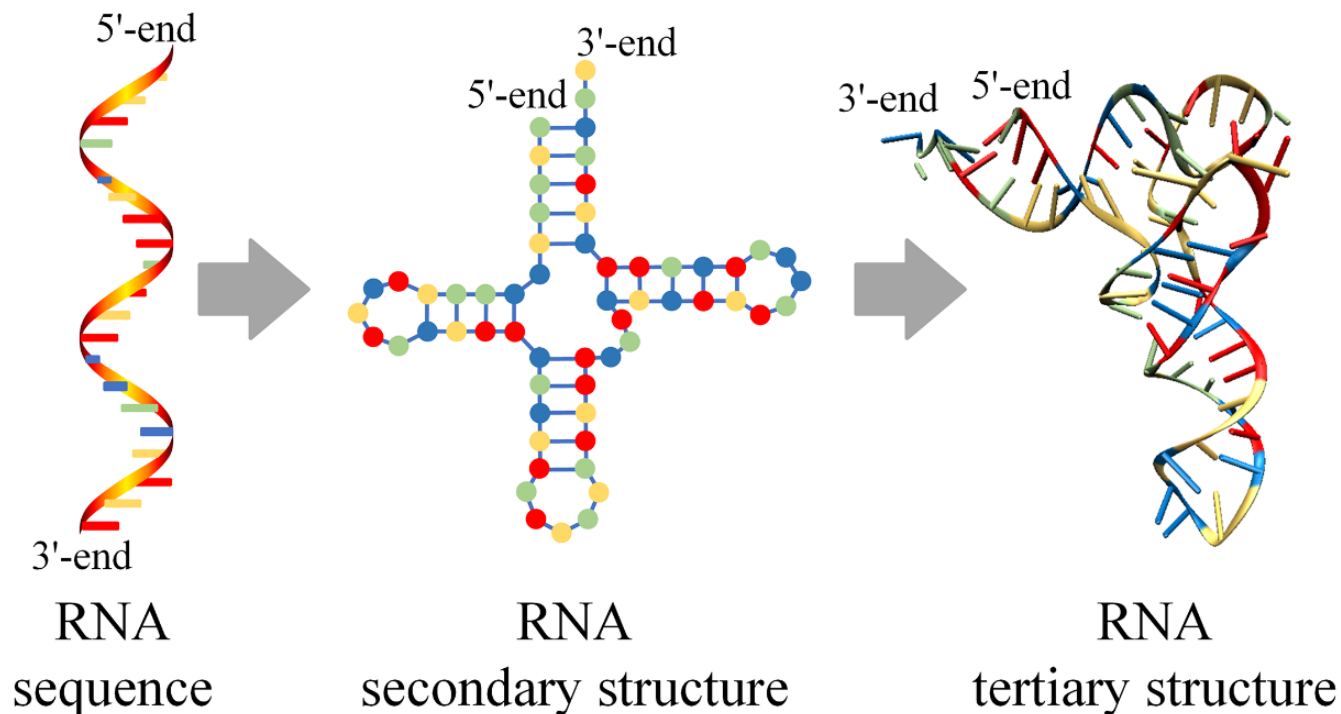


Nucleobases
of RNA

Bioinformatics

The RNA representation levels

RNA differently from DNA is a single strand molecule which can fold on himself creating complex structures → functional to biological tasks



Bioinformatics: isomiR-SEA C++ tool

BMC Bioinformatics

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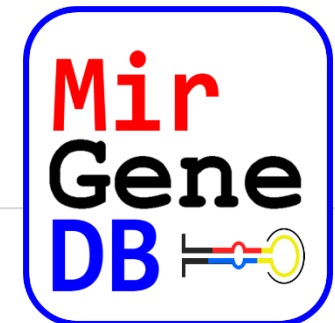
Software | [Open Access](#) | [Published: 31 March 2016](#)

isomiR-SEA: an RNA-Seq analysis tool for miRNAs/isomiRs expression level profiling and miRNA-mRNA interaction sites evaluation

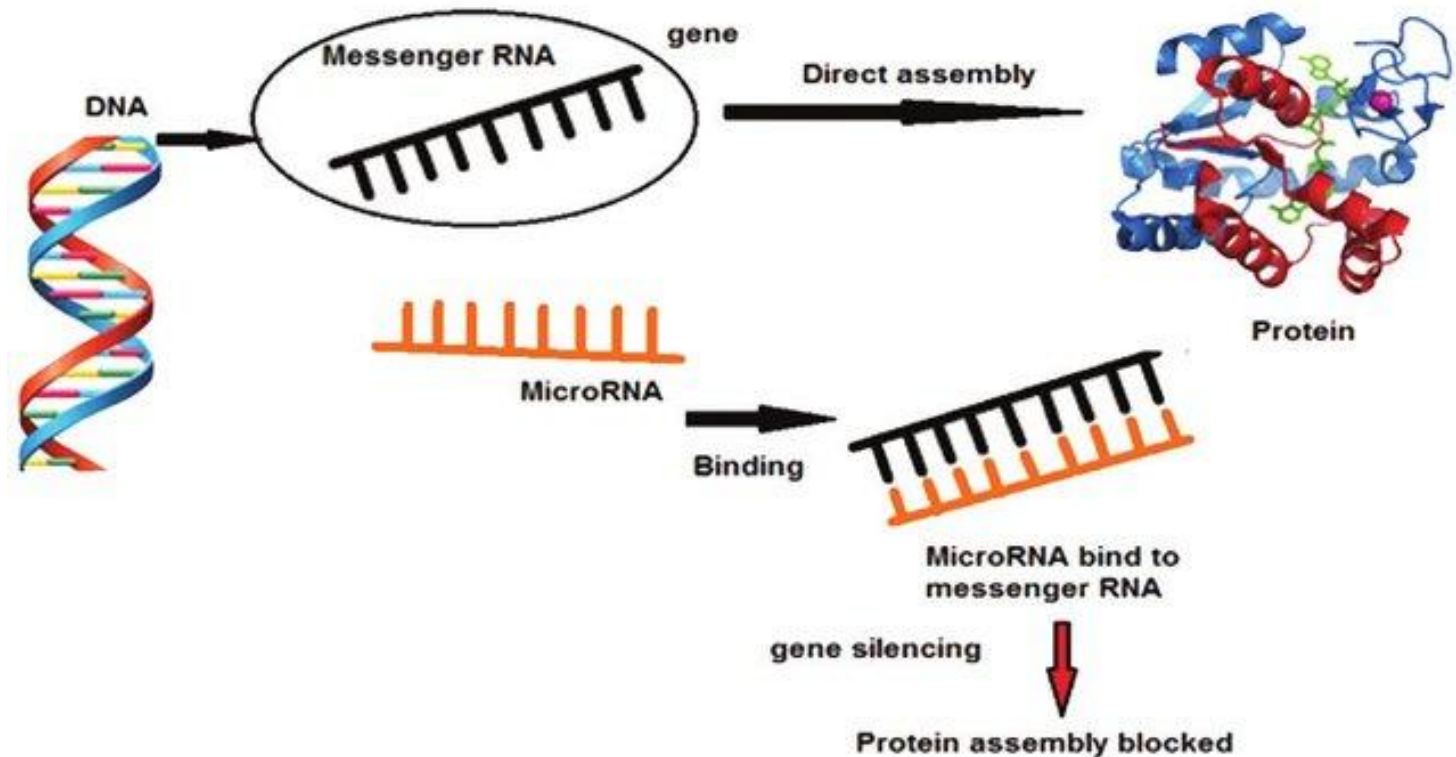
[Gianvito Urgese](#) , [Giulia Paciello](#), [Andrea Acquaviva](#) & [Elisa Ficarra](#)

[BMC Bioinformatics](#) **17**, Article number: 148 (2016) | [Cite this article](#)

3935 Accesses | **36** Citations | **12** Altmetric | [Metrics](#)

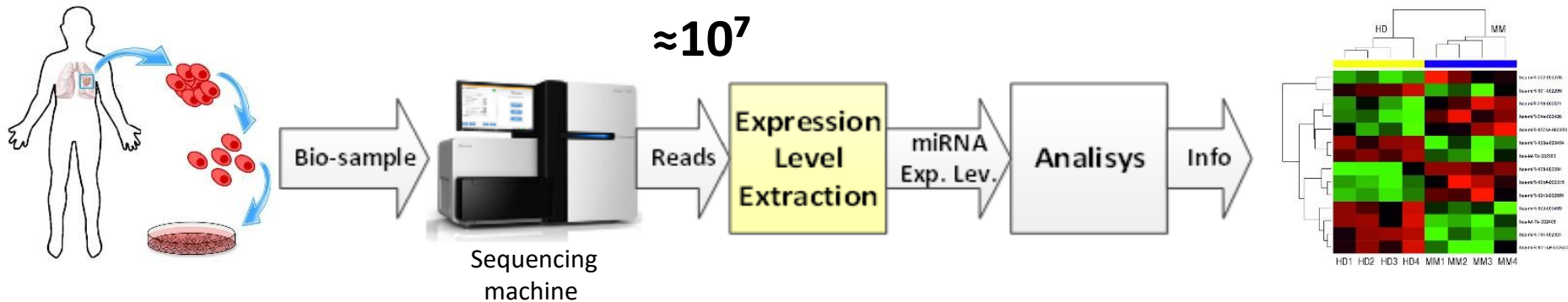


Bioinformatics: isomiR-SEA C++ tool



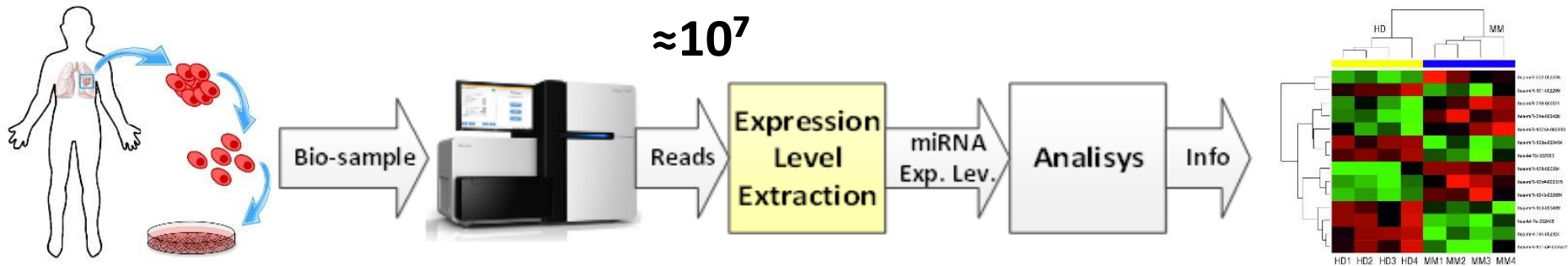
miRNAs are small RNA molecules capable to **downregulate** the expression of target **mRNA** through the binding with complementary mRNA sequences.

Bioinformatics: isomiR-SEA C++ tool



The analysis of miRNAs expression levels used to identify biomarkers into biological samples → specific patterns can be associated to some disease subtypes.

Bioinformatics: isomiR-SEA C++ tool

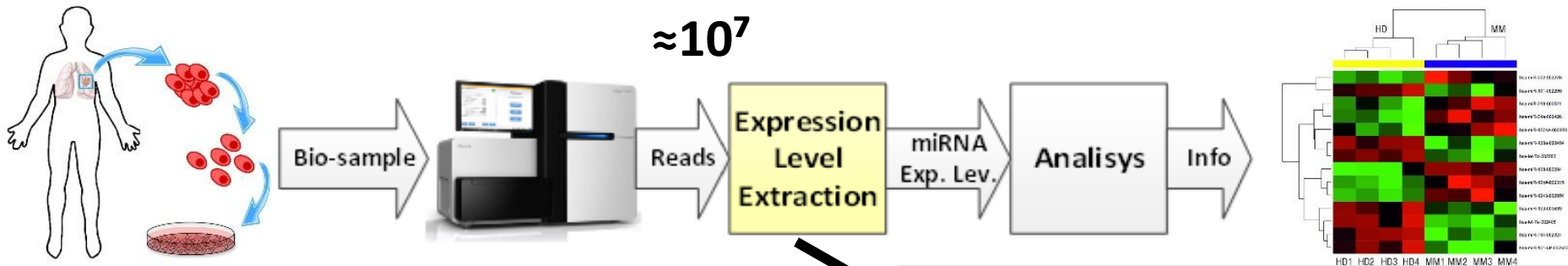


Tools reliability lack

CAUSE: Adaption of general-purpose alignment algorithms that does not take into account miRNA-specific evidences.

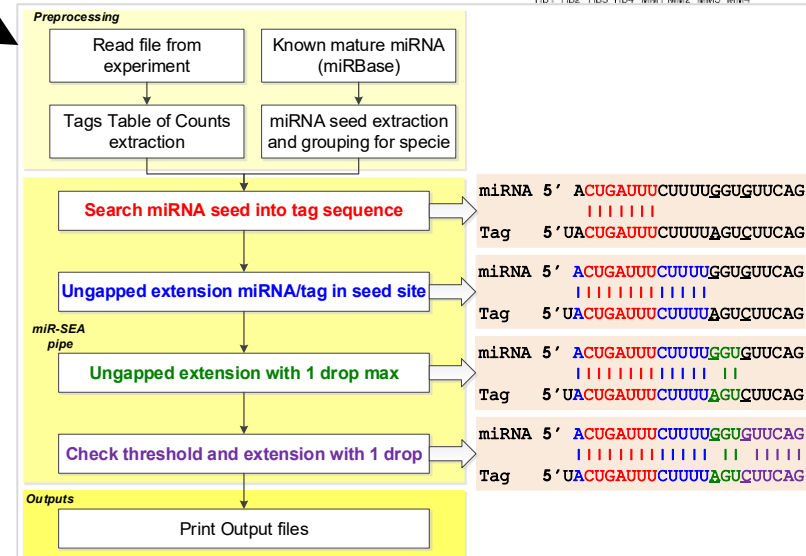
CONSEQUENCE: unreliable extracted expression levels

Bioinformatics: isomiR-SEA C++ tool

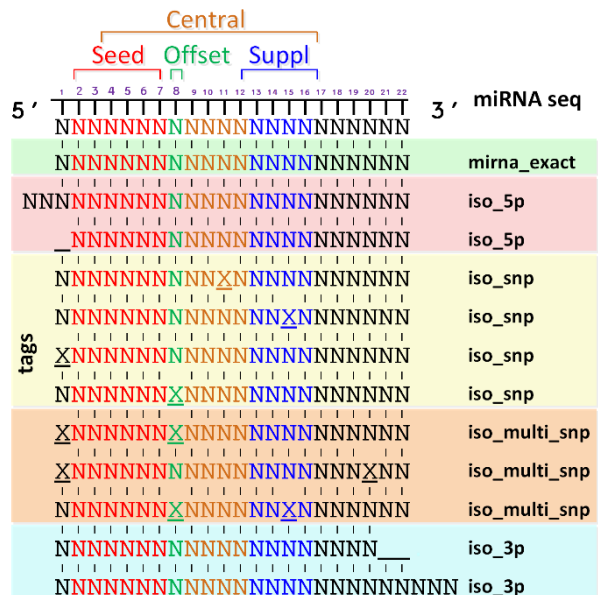
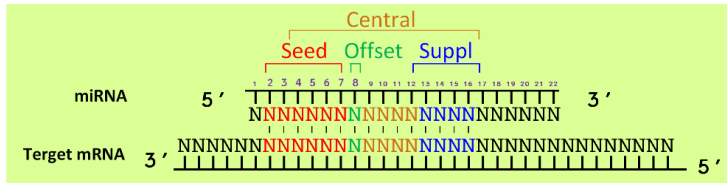


isomiR-SEA features

1. Alignment algorithm based on miRNA-specific evidences
2. Fast and accurate evaluation using the miRNA seed match identification phase
3. Early discard of tags inconsistent with miRNA structure
4. Pipeline easy to customize



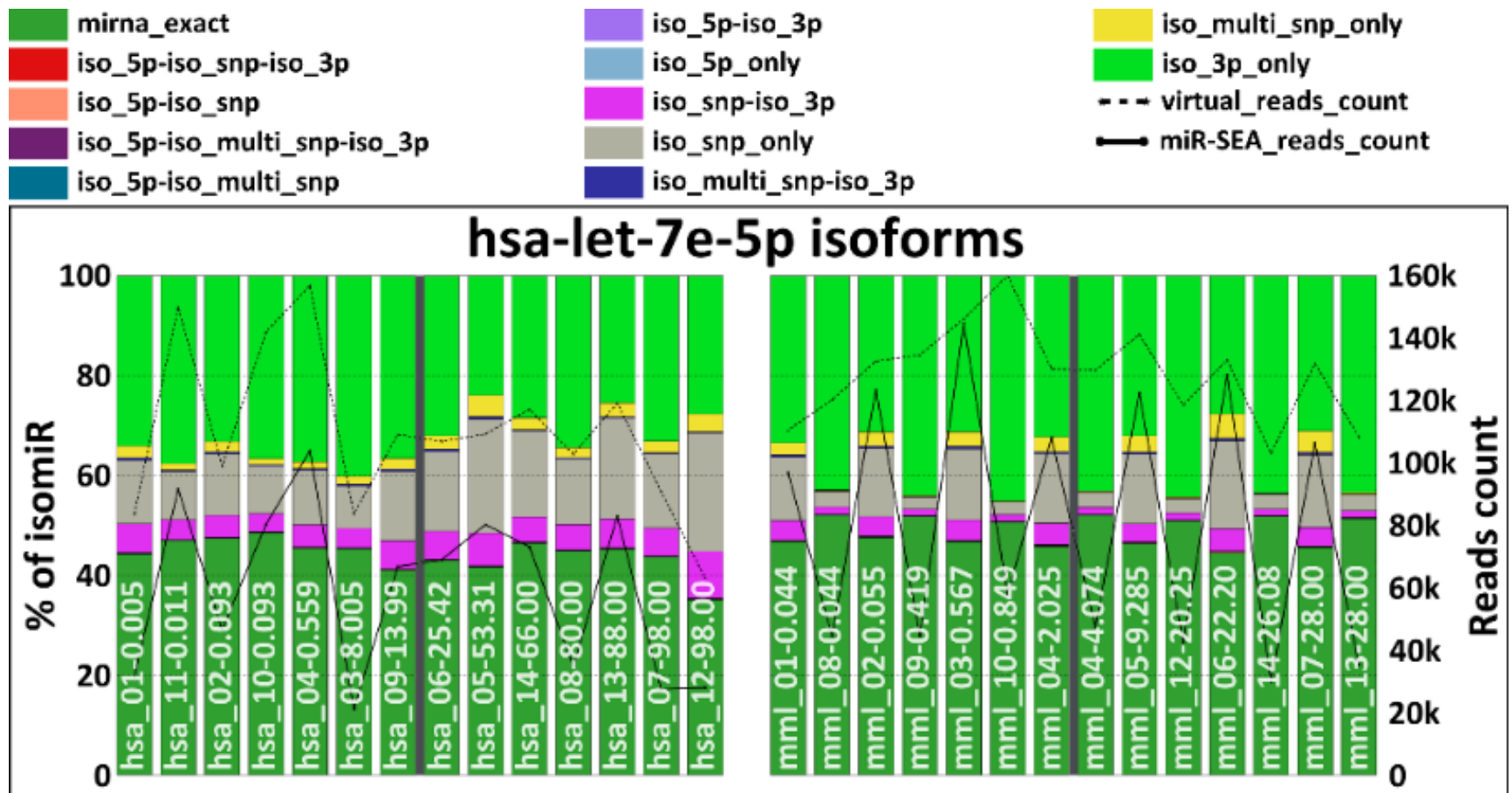
Bioinformatics: isomiR-SEA C++ tool



miRNA specific alignment algorithm

- Search the seed (seven nucleotide word)
- Ungapped extensions procedure
- Assignment of the read to the most similar sequence in the DB
- Identification of isoforms
- Annotation of all the important alignment info like the interaction site conservation

Bioinformatics: isomiR-SEA C++ tool



Bioinformatics: TCGA cancer dataset

? Attention GDC Users: The new date for the GDC User Support Webinar, Processing Whole Genome, Methylation, and Copy Number Data Types at the GDC, is June 6, 2022 at 2PM ET. You can register for the webinar [here](#). Dismiss ✕

NIH NATIONAL CANCER INSTITUTE GDC Data Portal

Home Projects Exploration Analysis Repository

Quick Search Manage Sets Login Cart 1 GDC Apps

Harmonized Cancer Datasets

Genomic Data Commons Data Portal

Get Started by Exploring:

Projects Exploration Analysis Repository

Q e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2

Data Portal Summary [Data Release 33.1 - May 31, 2022](#)

PROJECTS 72	PRIMARY SITES 67	CASES 85.552
FILES 828.104	GENES 21.759	MUTATIONS 2.683.650

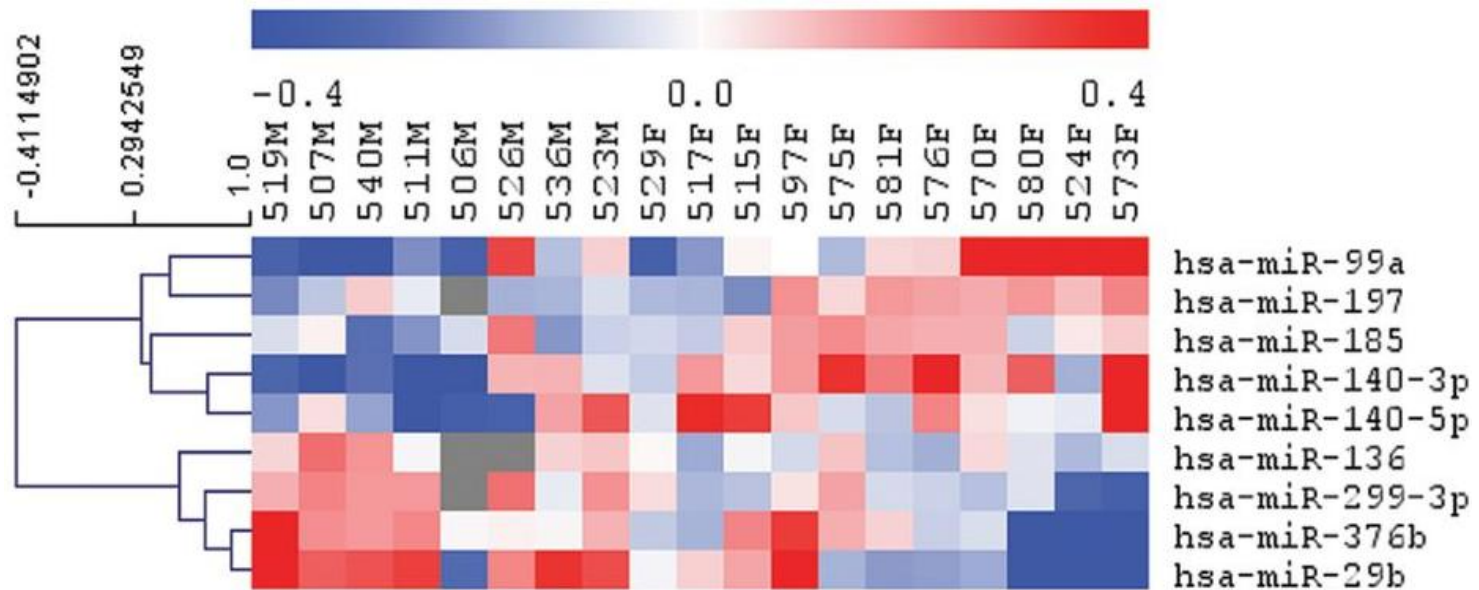
Cases by Major Primary Site

Adrenal Gland	0.1
Bile Duct	0.1
Bladder	0.1
Bone	0.1
Bone Marrow	0.1
Brain	0.1
Breast	9.5
Cervix	0.1
Colorectal	8.5
Esophagus	0.1
Eye	0.1
Head and Neck	0.1
Kidney	0.1
Liver	0.1
Lung	11.5
Lymph Nodes	0.1
Nervous System	0.1
Ovary	0.1
Pancreas	0.1
Pleura	0.1
Prostate	3.5
Skin	0.1
Soft Tissue	0.1
Stomach	0.1
Testis	0.1
Thymus	0.1
Thyroid	0.1
Uterus	0.1

ML in applications

Bioinformatics: Deep Learning Applications

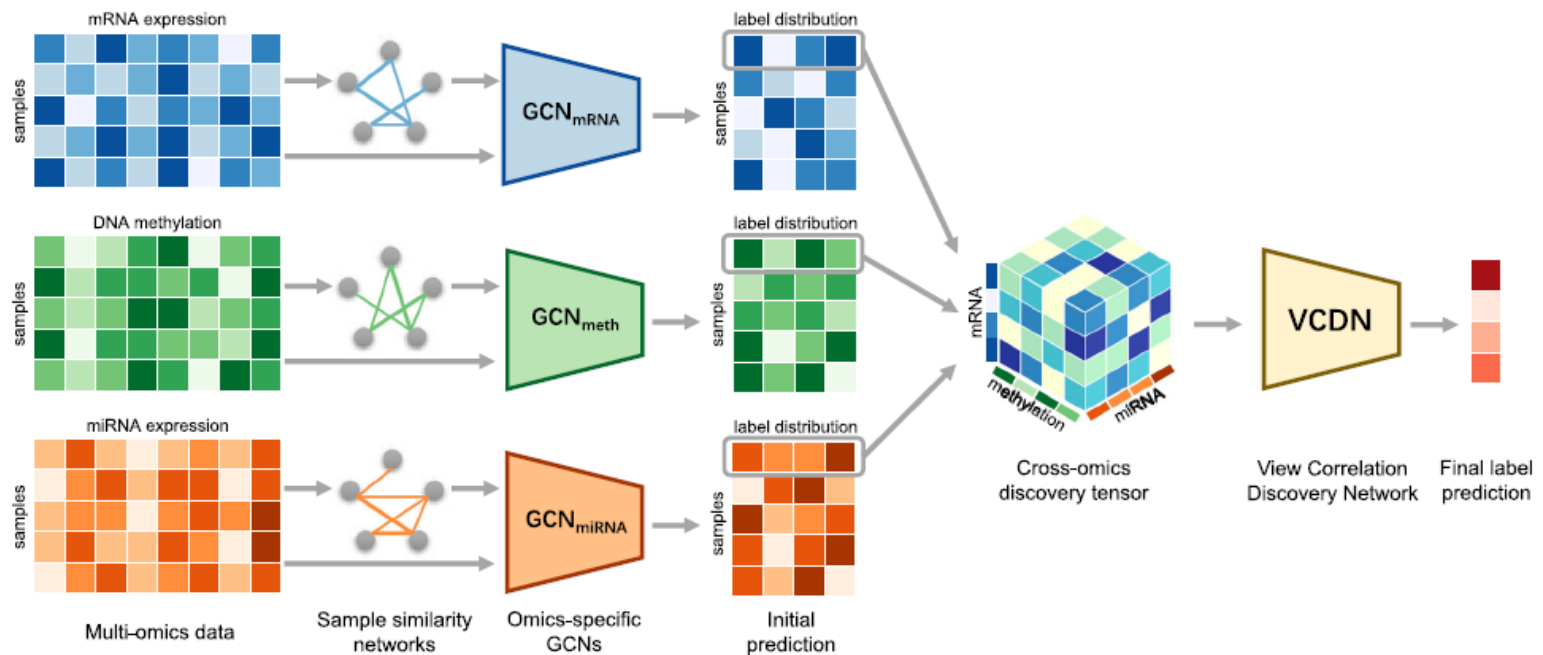
Application 1: Multi-omics-specific learning



miRNA expression level matrix

Bioinformatics: Deep Learning Applications

Application 2: Multi-omics-specific learning



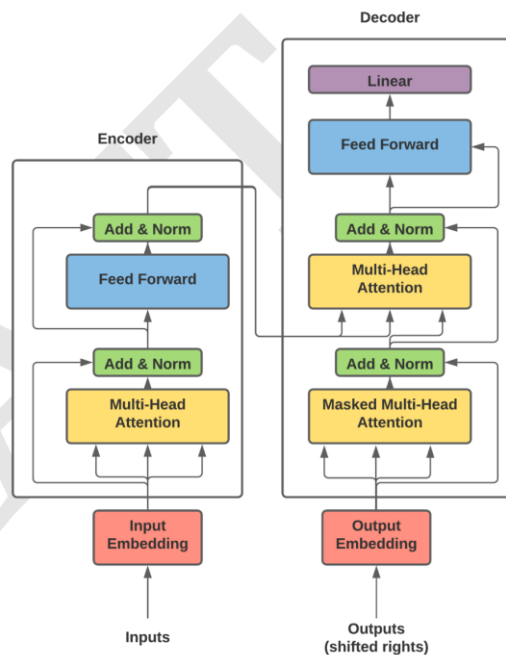
MOGONET combines Graph CNN for multi-omics-specific learning and VCDN for multi-omics integration.

Wang T. et al. «MOGONET integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification» 2021

Bioinformatics: Deep Learning Applications

Application 3: Multi-omics-specific learning

Cancers	CNVs	DNA methylation	miRNA	mRNA	protein
ACC		✓		✓	
BLCA		✓		✓	
BRCA			✓	✓	
CESC		✓	✓	✓	
CHOL		✓	✓	✓	
COAD			✓	✓	
DLBC		✓	✓	✓	
ESCA		✓	✓	✓	
GBM		✓	✓	✓	✓
HNSC		✓	✓	✓	
KICH		✓	✓	✓	
KIRC		✓	✓	✓	
KIRP		✓	✓	✓	
LAML		✓	✓	✓	
LGG		✓	✓	✓	✓
LIHC		✓	✓	✓	
LUAD		✓	✓	✓	
LUSC		✓	✓	✓	



	acc	prec	rec	f1_score
OmiVAE	97.49	-	-	97.5
OmiEmbed	97.71	-	-	96.83
XOmiVAE	-	-	-	90
GeneTransformer(8-Head)	-	96.02	95.61	95.64
MOT(1)	96.74	96.97	96.74	96.74
MOT(2)	97.30	97.48	97.30	97.33
MOT(3)	96.5	96.75	96.5	96.54
MOT(4)	98.4	98.50	98.4	98.37

MOT: a Multi-Omics Transformer for multiclass classification tumour types predictions

<https://doi.org/10.1101/2022.11.14.516459>

Bioinformatics Lab Instructions

 Drive

 Nuovo

 Priorità

 Il mio Drive

 Condivisi con me

 Recenti

 Speciali

 Cestino

 Archiviazione

Spazio di archiviazione utilizzato: 347 MB

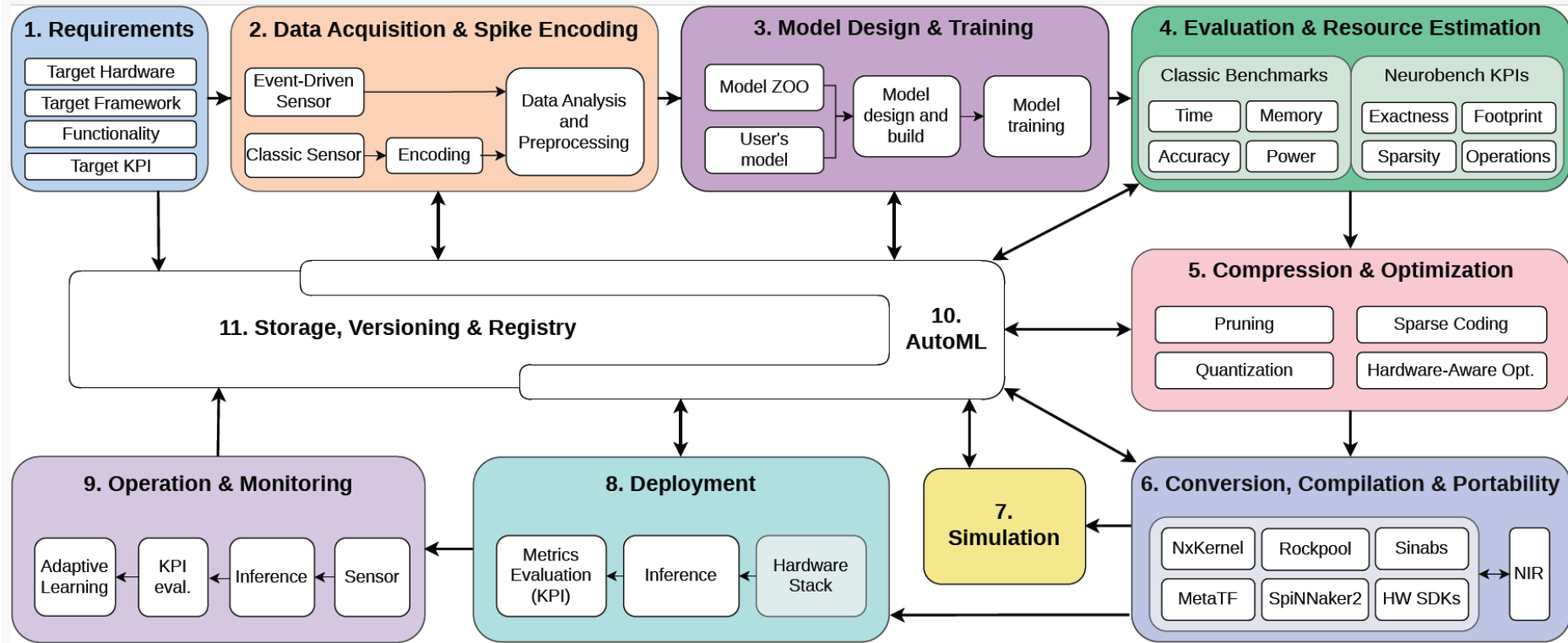
Cerca in Drive

Il mio Drive > ... > TCGA_miR_Dataset > txt > 

Nome 	Proprietario	Ultima modifica	Dimensioni file
 tcga_mir_label.csv 	io	4 mag 2022	3,8 MB
 tcga_mir_rpm.csv 	io	4 mag 2022	95,1 MB

<https://drive.google.com/drive/folders/1oWWeord8YYvtxlo2Pq2peyx7xOI-1Tmb?usp=sharing>

Neuromorphic MLOps



<https://ieeexplore.ieee.org/document/11393546/>

Bioinformatics Lab Instructions

tcga_mir_label.csv

Apri con Fogli Google

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
	file_id	file_name	cases_0_project_pro	file_size	case_id	demographic_days	demographic_ethnic	demographic_gender	demographic_race	demographic_vital	diagnoses_0_age_at	diagnoses_0_primary	disease_type	primary_site	project
1	5af40aa8-5713-465f-e0f44aa5-6938-435f	TARGET-RT		50652	54bacd1c-a6c0-5c2d-8b7b-54c0	167	not hispanic or latino	male	white	Dead		312	Malignant thabdo	Complex Mixed and Kidney	TARGE
2	6b4e065a-284a-40b1-1294f871-d0f7-4e3b	TARGET-RT		50749	6ed0b780-1b87-54c0-8b7b-54c0	177	not hispanic or latino	female	white	Dead		343	Malignant thabdo	Complex Mixed and Liver and Intrahepat	TARGE
3	dd436a3e-0939-4cfe-f01d85e9-ee5a-4d9f	TARGET-RT					male	white	Alive			761	Malignant thabdo	Complex Mixed and Lip	TARGE
4	2fec0342-019f-4873-2727a1d3-47a0-40e1	TARGET-RT					male	black or african ame	Alive			146	Malignant thabdo	Complex Mixed and Kidney	TARGE
5	c6bcd206-baca-4a6a-a290e5b1-1a1a-4a6a	TARGET-RT					male	white	Alive			141	Malignant thabdo	Complex Mixed and Lip	TARGE
6	33046a34-479b-4388-691fa711-4243-4511	TARGET-RT					male	other	Dead			64	Malignant thabdo	Complex Mixed and Kidney	TARGE
7	f334aca2-be9a-469c-5a9380cc-6ffc-41c9	TARGET-RT					male	white	Dead			286	Malignant thabdo	Complex Mixed and Kidney	TARGE
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11	4ab1f5e7-fb37-4915-9708192b-09e1-428f	TARGET-RT		50577	8afe6e5-753f-5272-b9e1-b491d883316a		Unknown	male	white	Alive		569	Malignant thabdo	Complex Mixed and Kidney	TARGE
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23	bd064faf-48c4-4966-313a33d4-8a57-499a	TARGET-RT		50294	5cdd05ea-5285-50b1-8b7b-54c0	102	not hispanic or latino	male	white	Dead		361	Malignant thabdo	Complex Mixed and Kidney	TARGE
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27	14f233d4-924a-417a-a6566173-3516-4e2f	TARGET-RT		50255	b34991f5-fa49-5b35-8e1b-95794d23ca12		hispanic or latino	female	white	Alive		214	Malignant thabdo	Complex Mixed and Kidney	TARGE

<https://drive.google.com/drive/folders/1oWWeord8YYvtxlo2Pq2peyx7xOI-1Tmb?usp=sharing>

Bioinformatics Lab Instructions

tcga_mir_label.csv

Apri con Fogli Google

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
	file_id	file_name	cases_0_project_pro	file_size	case_id	demographic_days	demographic_ethnic	demographic_gender	demographic_race	demographic_vital	diagnoses_0_age_at	diagnoses_0_primary	disease_type	primary_site	project
1	5af0aa8-5713-4655-e0f44aa5-6938-435f	1294f871-d0f7-4e3b	TARGET-RT	50652	54bacd1c-a6c0-5c2	167	not hispanic or latino	male	white	Dead	312	Malignant thabdo	Complex Mixed and	Kidney	TARGE
2	6b4e065a-284a-40b1-1294f871-d0f7-4e3b	1294f871-d0f7-4e3b	TARGET-RT	50749	6ed0b780-1b87-54c	177	not hispanic or latino	female	white	Dead	343	Malignant thabdo	Complex Mixed and	Liver and Intrahepat	TARGE
3	dd436a3e-0939-4cfc-f01d85e9-ee5a-4d9f	1294f871-d0f7-4e3b	TARGET-RT	50750	30ba94a6-bd7a-5d42-add6-c007b37888d		not hispanic or latino	female	white	Alive	761	Malignant thabdo	Complex Mixed and	Lip	TARGE
4	2fec0342-019f-4873-2727a1d3-47a0-40e	1294f871-d0f7-4e3b	TARGET-RT	50841	26079d5f-b061-5a39-968c-199e017cb88c		not hispanic or latino	male	black or african ame	Alive	146	Malignant thabdo	Complex Mixed and	Kidney	TARGE
5	c6bcd206-baca-4a68-a290e5b4-bb46-4f3	1294f871-d0f7-4e3b	TARGET-RT	50818	153					Alive	141	Malignant thabdo	Complex Mixed and	Lip	TARGE
6	33046a34-479b-4388-691fa711-4243-451	1294f871-d0f7-4e3b	TARGET-RT	50818	153					Dead	64	Malignant thabdo	Complex Mixed and	Kidney	TARGE
7	f334aca2-be9a-469c-5a9380cc-6ffc-41c9	1294f871-d0f7-4e3b	TARGET-RT	50326	e4c					Dead	286	Malignant thabdo	Complex Mixed and	Kidney	TARGE
8	bc7dc5d-e0d9-4ea8-bf0c8d01-6d51-430	1294f871-d0f7-4e3b	TARGET-RT	50559	e4c					Dead	286	Malignant thabdo	Complex Mixed and	Kidney	TARGE
9	3dac27d5-2b45-4b22-62d5a99f-7922-424	1294f871-d0f7-4e3b	TARGET-RT	50564	c1d					Dead	205	Malignant thabdo	Complex Mixed and	Kidney	TARGE
10	40d5bdc4-85ba-46e7-7d78d160-a2e3-499	1294f871-d0f7-4e3b	TARGET-RT	50593	51e					Dead	948	Malignant thabdo	Complex Mixed and	Kidney	TARGE
11	4ab1f5e7-b37-4915-9708192b-09e1-428	1294f871-d0f7-4e3b	TARGET-RT	50577	8afe6e5-753f-5272-b9e1-b491d883316a		Unknown	male	white	Alive	569	Malignant thabdo	Complex Mixed and	Kidney	TARGE
12	db6ce0d4-14c1-4611-4997f013-bcc5-451	1294f871-d0f7-4e3b	TARGET-RT	50572	36231648-a505-504	286	hispanic or latino	male	Unknown	Dead	1085	Malignant thabdo	Complex Mixed and	Kidney	TARGE
13	73d19446-b48e-4a41-22f5ebcb-1e4c-471	1294f871-d0f7-4e3b	TARGET-RT	50642	2e3c665d-8b99-503	120	Unknown	female	Unknown	Dead	541	Malignant thabdo	Complex Mixed and	Kidney	TARGE
14	64df9b95-7e21-4c43-fda4a49-db8d-41e2	1294f871-d0f7-4e3b	TARGET-RT	50286	21eae4ec-c6c6-5b7e-9d09-3135e435bca		Unknown	male	white	Unknown	290	Malignant thabdo	Complex Mixed and	Kidney	TARGE
15	cb6a9160-441e-4fc7-14f8b381-f10f-402a	1294f871-d0f7-4e3b	TARGET-RT	50512	a8bd90de-af34-5b8e-9a69-11befe50fb81		not hispanic or latino	male	white	Alive	352	Malignant thabdo	Complex Mixed and	Kidney	TARGE
16	c2a8d4f3-ed9b-4454-19a8b343-13c3-4e9	1294f871-d0f7-4e3b	TARGET-RT	50648	c8c222b3-b435-54c	86	not hispanic or latino	female	white	Dead	63	Malignant thabdo	Complex Mixed and	Kidney	TARGE
17	40bce46-14ea-48e7-7211caed-aa1a-43e	1294f871-d0f7-4e3b	TARGET-RT	50574	3479c5b5-51ce-57e	198	not hispanic or latino	male	white	Dead	1141	Malignant thabdo	Complex Mixed and	Kidney	TARGE
18	99271771-ebed-4655-575d8181-b296-407	1294f871-d0f7-4e3b	TARGET-RT	50661	ef42d20f-5f3e-58c6	48	Unknown	male	black or african ame	Dead	592	Malignant thabdo	Complex Mixed and	Kidney	TARGE
19	094dce50-0ff0-43d1-c9c390b9-0670-4b2	1294f871-d0f7-4e3b	TARGET-RT	50631	3e859917-be72-55d5-afc0-dffa8e6cd12e		not hispanic or latino	male	white	Alive	147	Malignant thabdo	Complex Mixed and	Kidney	TARGE
20	91319dd2-c7b0-40a1-b3611511-8cef-409	1294f871-d0f7-4e3b	TARGET-RT	50423	fc886b71-cf65-5853	1639	not hispanic or latino	female	Unknown	Dead	138	Malignant thabdo	Complex Mixed and	Kidney	TARGE
21	0b593325-ea7e-4955-878e9476-5109-4e5	1294f871-d0f7-4e3b	TARGET-RT	50574	1e1973ab-60fa-5bd1-8a97-47b4e18482a3		not hispanic or latino	male	white	Alive	1038	Malignant thabdo	Complex Mixed and	Kidney	TARGE
22	a83dd106-bf29-4257-ce0dfac6-ae00-41c	1294f871-d0f7-4e3b	TARGET-RT	50810	33c9c596-9412-54ff-8902-5b8060fded60		not hispanic or latino	female	white	Alive	239	Malignant thabdo	Complex Mixed and	Kidney	TARGE
23	bd064faf-48c4-4966-313a33d4-8a57-49a	1294f871-d0f7-4e3b	TARGET-RT	50294	5cdd05ea-5285-50b	102	not hispanic or latino	male	white	Dead	361	Malignant thabdo	Complex Mixed and	Kidney	TARGE
24	73749277-0455-422b-a38b84e-9f6a-4b6	1294f871-d0f7-4e3b	TARGET-RT	50711	5cdd05ea-5285-50b	102	not hispanic or latino	male	white	Dead	361	Malignant thabdo	Complex Mixed and	Kidney	TARGE
25	28b2a80d-9a84-43a1-e494428e-6999-40f	1294f871-d0f7-4e3b	TARGET-RT	50509	9d6b9b3b-04eb-54c	386	Unknown	female	Unknown	Dead	1466	Malignant thabdo	Complex Mixed and	Kidney	TARGE
26	60fd20f-2ba0-4f76-013306dd-77fa-48a	1294f871-d0f7-4e3b	TARGET-RT	50615	ea300498-806d-56cf-9cca-adb64f869e7b		hispanic or latino	male	white	Alive	220	Malignant thabdo	Complex Mixed and	Kidney	TARGE
27	14f233d4-924a-417e-a6566173-3516-4e2	1294f871-d0f7-4e3b	TARGET-RT	50255	b34991f5-fa49-5b35-8e1b-95794d23ca12		hispanic or latino	female	white	Alive	214	Malignant thabdo	Complex Mixed and	Kidney	TARGE

Tumour classes

<https://drive.google.com/drive/folders/1oWWeord8YYvtxlo2Pq2peyx7xOI-1Tmb?usp=sharing>

Bioinformatics Lab Instructions

← tcga_mir_label.csv

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
	file_id	file_name	cases_0_project_pro	file_size	case_id	demographic_days	demographic_ethnic	demographic_gender	demographic_race	demographic_vital	diagnoses_0_age_at	diagnoses_0_primary	disease_type	primary_site	project
1	5af40aa8-5713-465c-e0f44a5-6938-435f	TARGET-RT			50652	54bacd1c-a6c0-5c2	167	not hispanic or latino	male	white	Dead	312	Malignant thabdo	Complex Mixed and Kidney	TARGE
2	6b4e065a-284a-40b1-1294f871-d0f7-4e3b	TARGET-RT			50749	6ed0b780-1b87-54c	177	not hispanic or latino	female	white	Dead	343	Malignant thabdo	Complex Mixed and Liver and Intrahepat	TARGE
3	dd436a3e-0939-4cft-f01d85e9-ee5a-4d9f	TARGET-RT			50750	30ba94a6-bd7a-5d42-add6-c007b37888d		not hispanic or latino	female	white	Alive	761	Malignant thabdo	Complex Mixed and Lip	TARGE
4	2fec0342-019f-4873-2727a1d3-47a0-40e1	TARGET-RT			50841	26079d5f-b061-5a39-968c-199e017cb88c		not hispanic or latino	male	black or african ame	Alive	146	Malignant thabdo	Complex Mixed and Kidney	TARGE
5	c6bcd206-baca-4a6a-a290e5b4-bb46-4f33	TARGET-RT			50818	153c5c30-41d5-5369-873f-d086ead699a		not hispanic or latino	female	white	Alive				TARGE
6	33046a34-479b-438-691fa711-4243-4511	TARGET-RT			50617	e2ac740d-a9e9-58c	140	not hispanic or latino	female	other					TARGE
7	f33aca2-ba9a-469c-5a9380cc-6ffc-41c9	TARGET-RT			50326	e4c16a18-930e-5c7	187	not hispanic or latino	male	white	Dead				TARGE
8	bc7dcd5d-a0d9-4ea-bf0c8d01-6d51-4307	TARGET-RT			50559	e4c16a18-930e-5c7	187	not hispanic or latino	male	white	Dead				TARGE
9	3dac27d5-2b45-4b2-62d5a99f-7922-4241	TARGET-RT			50564	c1d975a0-49ca-5c95	336	not hispanic or latino	male	white	Dead				TARGE
10	40d5bdc4-85ba-46e-7d78d160-a2e3-499	TARGET-RT			50593	51edacf9-41a8-594C	243	not hispanic or latino	female	white	Dead				TARGE
11	4ab1f5e7-fb37-4915-9706192b-09e1-428	TARGET-RT			50577	8afe6e5-753f-5272-b9e1-b491d883316a	Unknown	male	white	Alive	569	Malignant thabdo	Complex Mixed and Kidney	TARGE	
12	db6ce0d4-14c1-461-4997f013-bcc5-4511	TARGET-RT			50572	36231648-a505-504	286	hispanic or latino	male	Unknown	Dead	1085	Malignant thabdo	Complex Mixed and Kidney	TARGE
13	73d19446-b48e-4a4-22f5ebcb-1e4c-4715	TARGET-RT			50642	2e3c685d-8b99-503	120	Unknown	female	Unknown	Dead	541	Malignant thabdo	Complex Mixed and Kidney	TARGE
14	64df9b95-7e21-4c43-fda4a49-db8d-41e2	TARGET-RT			50286	21eae4ec-c6c6-5b7e-9d09-3135e435bca	Unknown	male	white	Unknown		290	Malignant thabdo	Complex Mixed and Kidney	TARGE
15	cb6a9160-441e-4fc7-14f8b381-f10f-402a	TARGET-RT			50512	a8bd90de-af34-5b8e-9a69-11bef50fb81		not hispanic or latino	male	white	Alive	352	Malignant thabdo	Complex Mixed and Kidney	TARGE
16	c2a8d4f3-ed9b-4454-19a8b343-13c3-4e9	TARGET-RT			50648	c8c222b3-b435-54c	86	not hispanic or latino	female	white	Dead	63	Malignant thabdo	Complex Mixed and Kidney	TARGE
17	40bce46-14ea-48e-7211caed-aa1a-43d	TARGET-RT			50574	3479c5b5-51ce-57e	198	not hispanic or latino	male	white	Dead	1141	Malignant thabdo	Complex Mixed and Kidney	TARGE
18	99271771-ebed-465-575d8181-b296-407	TARGET-RT			50661	ef42d20f-5f3e-58c6	48	Unknown	male	black or african ame	Dead	592	Malignant thabdo	Complex Mixed and Kidney	TARGE
19	094dce50-0ff0-43d1-c9c390b9-0670-4b2	TARGET-RT			50631	3e859917-bef2-55d5-afc0-dffa8e6cd12e		not hispanic or latino	male	white	Alive	147	Malignant thabdo	Complex Mixed and Kidney	TARGE
20	91319dd2-c7b0-40a-b3611511-8cef-4095	TARGET-RT			50423	fc8b6b71-cf65-5853	1639	not hispanic or latino	female	Unknown	Dead	138	Malignant thabdo	Complex Mixed and Kidney	TARGE
21	0b593325-ea7e-495-878e9476-5109-4e5	TARGET-RT			50574	1e1973ab-60fa-5bd1-8a97-47b4e18482a3		not hispanic or latino	male	white	Alive	1038	Malignant thabdo	Complex Mixed and Kidney	TARGE
22	a83dd106-bf29-425f-ce0dfac6-ae00-41c4	TARGET-RT			50810	33c9c596-9412-54ff-8902-5b8060fed60		not hispanic or latino	female	white	Alive	239	Malignant thabdo	Complex Mixed and Kidney	TARGE
23	bd064faf-48c4-496c-313a33d4-8a57-49a	TARGET-RT			50294	5cdd05ea-5285-50b	102	not hispanic or latino	male	white	Dead	361	Malignant thabdo	Complex Mixed and Kidney	TARGE
24	73749277-0455-422-ba38b84e-9f6a-4b63	TARGET-RT			50711	5cdd05ea-5285-50b	102	not hispanic or latino	male	white	Dead	361	Malignant thabdo	Complex Mixed and Kidney	TARGE
25	28b2a80d-9a84-43a-e494428e-6999-40f	TARGET-RT			50509	9d6b9b3b-04eb-54c	386	Unknown	female	Unknown	Dead	1466	Malignant thabdo	Complex Mixed and Kidney	TARGE
26	60fd20f-2ba0-4f76-013306dd-77fa-48a5	TARGET-RT			50615	ea300498-806d-56cf-9cca-adb64f869e7b		hispanic or latino	male	white	Alive	220	Malignant thabdo	Complex Mixed and Kidney	TARGE
27	14f233d4-924a-417e-a6566173-3516-4e2	TARGET-RT			50255	b34991f5-fa49-5b35-8e1b-95794d23ca12		hispanic or latino	female	white	Alive	214	Malignant thabdo	Complex Mixed and Kidney	TARGE

Sex classes

<https://drive.google.com/drive/folders/1oWWeord8YYvtxlo2Pq2peyx7xOI-1Tmb?usp=sharing>

Bioinformatics Lab Instructions

tcga_mir_rpm.csv

Table "default" - Rows: 11661 | Spec - Columns: 1882 | Properties | Flow Variables

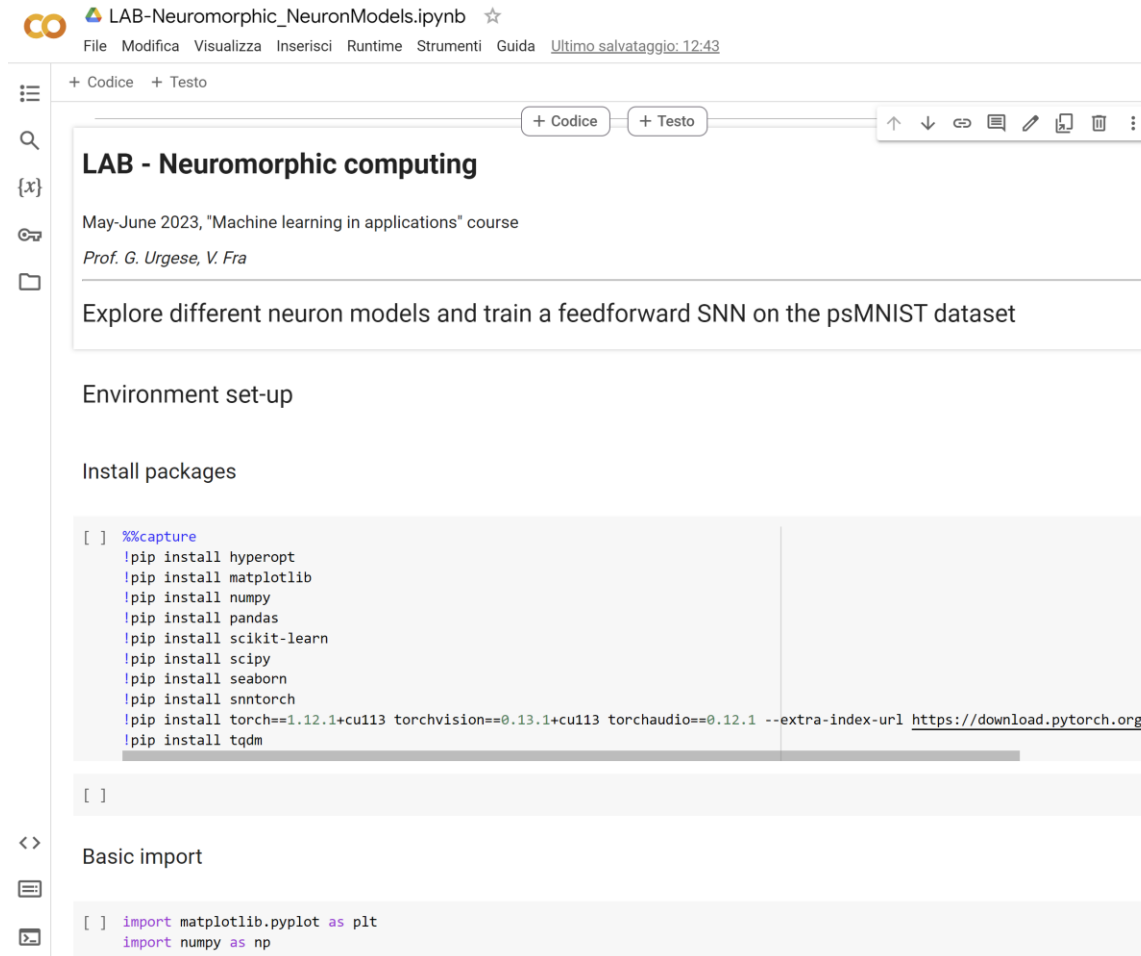
Row ID	D hsa-mir...	I hsa-mir...	I hsa-mir...	I hsa-mir...	I hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	D hsa-mir...	S file_id
Row11624	0	0	0	0	0	4.782	0	0.869	2.826	0	1.521	42.383	1,129.999	3de2e5b4-fdde-4b37-91e7-21a72ccc97f9
Row11625	0	0	0	0	0	17.206	0	184.245	1.434	0	91.764	77.426	248.049	fc4feac9-580b-49c2-a96f-0022abf2299
Row11626	0	0	0	0	0	4.531	0	0.755	0.755	0	23.41	74.382	100.812	6ea3e80f-f23d-4ffc-9154-60ce3e2a8210
Row11627	0	0	0	0	0	6.723	0	115.539	2.49	0	21.415	45.568	554.289	16d2e3ba-c3f4-4e7d-a304-091e932d3459
Row11628	0	0	0	0	0	7.828	0	185.516	0.783	0	54.533	68.362	107.239	af88c93a-9c45-4cf9-b0ab-6a55c50694a7
Row11629	0	0	0	0	0	18.455	0.179	6.629	2.329	0	14.871	92.99	47.839	c6d8f8fc-02e1-4b5c-8d1f-589a084d5538
Row11630	0	0	0	0	0	26.009	0	266.129	1.04	0	15.606	65.752	393.888	622adda6-61d3-4488-a306-2b1ae53df65a
Row11631	0	0	0	0	0	11.452	0	106.5	1.145	0	111.081	77.871	151.161	e2045e4c-ab0c-4f76-a722-a03e52243bf6
Row11632	0	0	0	0	0	27.174	0	55.706	2.989	0	5.978	44.293	209.511	6cb1706e-6d10-4c34-931c-9d6013f75bd
Row11633	0	0	0	0	0	8.362	0	51.565	0	0	36.235	42.506	193.019	41cbfc51-4f09-49f0-8766-20daf6f9442f
Row11634	0	0	0	0	0	8.208	0.164	52.2	3.611	0	8.864	67.794	362.607	010c80c8-2f98-48b0-8320-826db6b8fc50
Row11635	0	0	0	0	0	6.176	0	133.962	3.325	0	35.628	62.231	140.613	0bbe2ff8-d631-414e-ae09-e4c459219a40
Row11636	0	0	0	0	0	16.665	0	84.047	2.898	0	36.227	52.892	469.506	0bff265a-8c84-48d9-b40b-99fa252c9b2a
Row11637	0	0	0	0	0	6.989	0	24.112	1.573	0	12.231	52.943	729.839	a4121a1e-607a-4223-b0f6-ca415832c0a4
Row11638	0	0	0	0	0	3.527	0	36.934	3.942	0	12.45	33.407	146.077	66aa1e0c-4b79-4307-a022-b3c3a9485625
Row11639	0	0	0	0	0	26.657	0.523	225.014	2.091	0	40.246	140.079	410.305	e6a52336-9088-4b4e-9ebf-a4e2d0f2e5ed
Row11640	0	0	0	0	0	6.612	0	352.999	0.509	0	104.272	75.788	106.307	5eba244a-3dda-43bb-9141-31656a10846d
Row11641	0	0	0	0	0	1.399	0	23.775	2.797	0	8.391	34.964	290.9	8f4184a6-245d-485a-86ce-074c815ccc2e
Row11642	0	0	0	0	0	10.721	0	1.031	2.474	0	4.948	40.824	906.272	54d2d5fb-2f06-4079-9001-e9e3130861b3
Row11643	0	0	0	0	0	19.441	0	8.729	1.19	0	22.416	97.402	163.212	1e9dc9ad-410e-44f2-aa55-b72b7b25527e
Row11644	0	0	0	0	0	10.921	0	134.695	19.882	0	15.962	53.766	22,254.065	7ee6eddf-9e75-4b2f-884f-7a17582ef015
Row11645	0	0	0	0	0	11.087	0	200.516	1.584	0	57.969	45.689	20,026.21	a4a80eba-ee29-4630-838a-2b5255a445e3
Row11646	0	0	0	0	0	6.898	0	0	0	0	0	0	17,643.529	902e7845-5e1f-475a-a2fd-c8ca68d25a6
Row11647	0	0	0	0	0	14.851	0	0	0	0	0	0	16,942.927	f250deb2-f14e-4f99-bbde-6dca36e15acf
Row11648	0	0	0	0	0	3.908	0	0	0	0	0	0	32,499.357	075d11f1-ec6c-43d7-a0ae-dac734fb4deb
Row11649	0	0	0	0	0	16.61	0	0	0	0	0	0	28,952.743	430a1d7c-3e97-4bcb-afe4-9e1331063a11
Row11650	0	0	0	0	0	10.18	0	0	0	0	0	0	9,943.518	fb1a02dd-bb94-4029-a514-bb37468a6604
Row11651	0	0	0	0	0	56.07	0	0	0	0	0	0	19,593.411	25b2ff7b-8256-4a32-a036-d4a463bf19e9
Row11652	0	0	0	0	0	22.637	0	0	0	0	0	0	250,641.384	36aed435-4e6f-46df-91ab-4dd009139520
Row11653	0	0	0	0	0	8.525	0	0	0	0	0	0	24,845.544	a2735168-4418-4a32-bf84-d86720f26dff
Row11654	0	0	0	0	0	7.524	0	128.412	2.006	0	8.026	52.167	339.088	76d399d9-79b9-4a6c-a1fb-48bedc051f6
Row11655	0	0	0	0	0	12.161	0	66.525	8.584	0	8.584	61.518	186.7	8e798832-aa7f-450f-878c-eb0d9878f651
Row11656	0	0	0	0	0	3.399	0.309	367.746	2.472	0	52.226	59.334	434.496	8f54427a-5780-458d-877d-cfc9df3153d
Row11657	0	0	0	0	0	5.107	0	226.395	0.851	0	58.726	85.962	165.115	ac15bae2-a450-4df5-8b6a-077391f83ae5
Row11658	0	0	0	0	0	7.788	0.159	192.644	1.907	0	70.414	85.831	259.084	c3847688-328e-49fc-8eee-6abaa875486b
Row11659	0	0	0	0	0	12.852	0	424.932	0.803	0	87.557	137.36	168.688	ce3a2d72-0ce9-4a6c-b311-cfe55598a8d
Row11660	0	0	0	0	0	9.687	0	400.023	4.726	0	81.517	93.094	424.123	5000b708-082b-4c9e-9632-94a5b7981700

miRNA expression levels

Link with entries in the dataset file

<https://drive.google.com/drive/folders/1oWWeord8YYvtxlo2Pq2peyx7xOI-1Tmb?usp=sharing>

Bioinformatics Lab Instructions



LAB-Neuromorphic_NeuronModels.ipynb ☆

File Modifica Visualizza Inserisci Runtime Strumenti Guida [Ultimo salvataggio: 12:43](#)

+ Codice + Testo

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LAB - Neuromorphic computing

May-June 2023, "Machine learning in applications" course

Prof. G. Urgese, V. Fra

Explore different neuron models and train a feedforward SNN on the psMNIST dataset

Environment set-up

Install packages

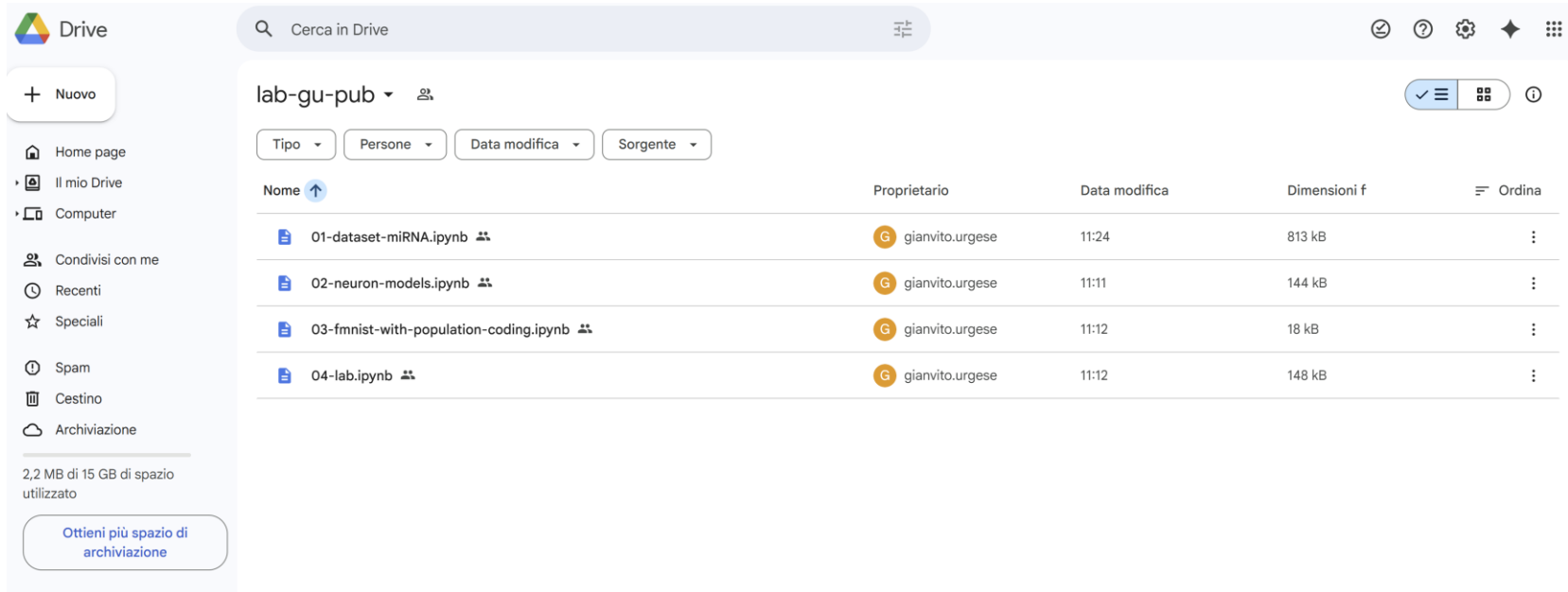
```
[ ] %%capture
!pip install hyperopt
!pip install matplotlib
!pip install numpy
!pip install pandas
!pip install scikit-learn
!pip install scipy
!pip install seaborn
!pip install snntorch
!pip install torch==1.12.1+cu113 torchvision==0.13.1+cu113 torchaudio==0.12.1 --extra-index-url https://download.pytorch.org/
!pip install tqdm
```

Basic import

```
[ ] import matplotlib.pyplot as plt
import numpy as np
```

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYp7vQ60-oYMzIJu34sbM?usp=sharing>

Bioinformatics Lab Instructions



The screenshot shows the Google Drive web interface. On the left is a sidebar with navigation options: 'Nuovo', 'Home page', 'Il mio Drive', 'Computer', 'Condivisi con me', 'Recenti', 'Speciali', 'Spam', 'Cestino', and 'Archiviazione'. The main area displays a folder named 'lab-gu-pub'. Below the folder name are filters for 'Tipo', 'Persone', 'Data modifica', and 'Sorgente'. A table lists the files in the folder:

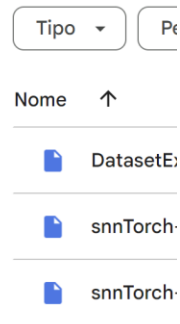
Nome	Proprietario	Data modifica	Dimensioni f	Ordina
01-dataset-miRNA.ipynb	gianvito.urgese	11:24	813 kB	
02-neuron-models.ipynb	gianvito.urgese	11:11	144 kB	
03-fmnist-with-population-coding.ipynb	gianvito.urgese	11:12	18 kB	
04-lab.ipynb	gianvito.urgese	11:12	148 kB	

At the bottom left of the sidebar, it indicates '2,2 MB di 15 GB di spazio utilizzato' and a button to 'Ottieni più spazio di archiviazione'.

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYP7vQ60-oYMzIJu34sbM?dmr=1&ec=wgc-drive-%5Bmodule%5D-goto>

Bioinformatics Lab Instructions

... > notebooks_2024 > lab-gu-pub ▾ 🔍



1. Download the full folder **lab_gu-pub**
2. Upload the folder in **your google account**
3. Run the notebooks on **google-colab** following the instructor
 - Run the notebooks on your system with GPU (Optional)

The full workflow will be used in the Bioinformatics Lab for classification of miRNA expression levels.

The task is to implement a classifier of tumour type (optional sex or some other attributes available in the label file).

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYP7vQ60-oYMzIJu34sbM?dmr=1&ec=wgc-drive-%5Bmodule%5D-goto>

Bioinformatics Lab Instructions

Dataset Exploration with techniques like PCA, t-SNE, UMAP and GNG, for a general overview of the datasets.

The screenshot shows a Google Drive interface. On the left is a sidebar with navigation options: 'Nuovo', 'Home page', 'Il mio Drive', 'Computer', 'Condivisi con me', 'Recenti', 'Speciali', 'Spam', 'Cestino', 'Archiviazione', and a storage status of '2,2 MB di 15 GB di spazio utilizzato' with a button to 'Ottieni più spazio di archiviazione'. The main area shows a folder named 'lab-gu-pub' with search filters for 'Tipo', 'Persone', 'Data modifica', and 'Sorgente'. Below these are four files:

Nome	Proprietario	Data modifica	Dimensioni f	Ordina
01-dataset-miRNA.ipynb	gianvito.urgese	11:24	813 kB	
02-neuron-models.ipynb	gianvito.urgese	11:11	144 kB	
03-fmnist-with-population-coding.ipynb	gianvito.urgese	11:12	18 kB	
04-lab.ipynb	gianvito.urgese	11:12	148 kB	

A callout box from the text above points to the first file, '01-dataset-miRNA.ipynb'.

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYP7vQ60-oYMzIJu34sbM?dmr=1&ec=wgc-drive-%5Bmodule%5D-goto>

Bioinformatics Lab Instructions

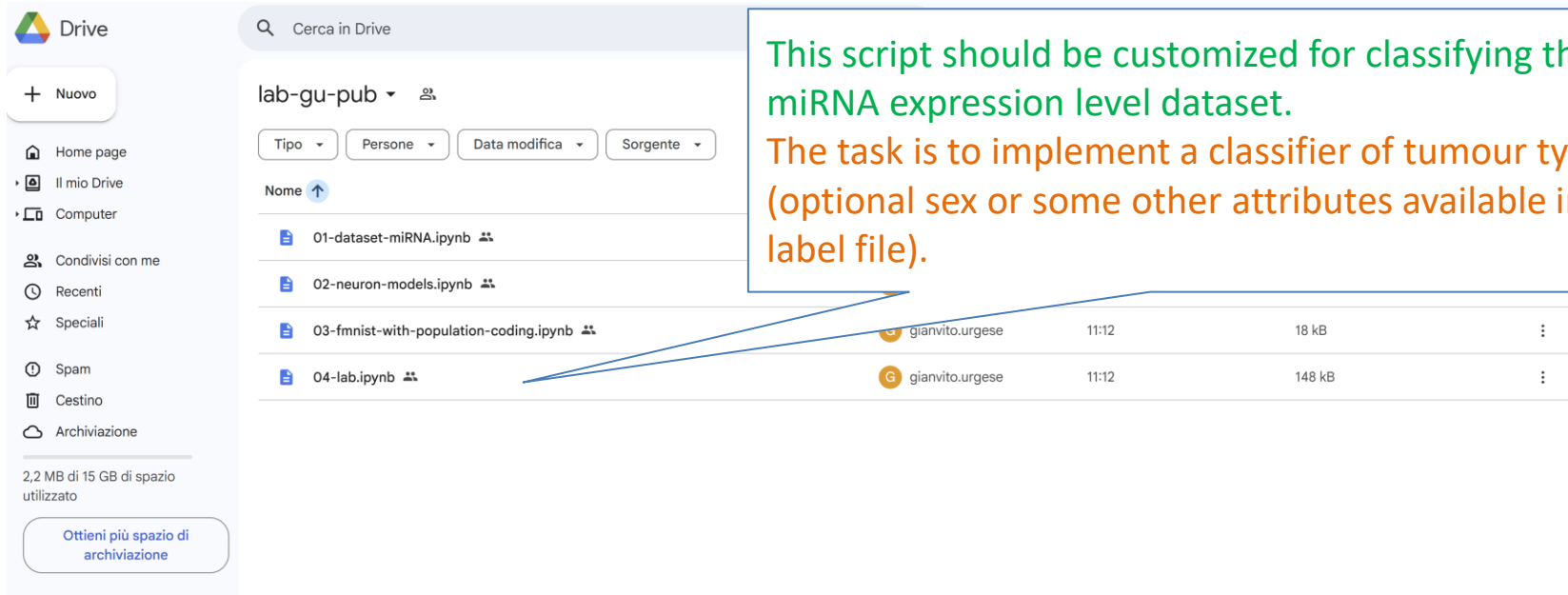
The screenshot shows the Google Drive interface. On the left is the sidebar with navigation options. The main area displays a folder named 'lab-gu-pub'. Below the folder name are filters for 'Tipo', 'Persone', 'Data modifica', and 'Sorgente'. A table lists the files in the folder:

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03-fmnist-with-population-coding.ipynb	gianvito.urgese	11:12	18 kB	
04-lab.ipynb	gianvito.urgese	11:12	148 kB	

A callout box points to the file '03-fmnist-with-population-coding.ipynb' with the text: Simple MNIST classification example with snnTorch

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYP7vQ60-oYMzIJu34sbM?dmr=1&ec=wgc-drive-%5Bmodule%5D-goto>

Bioinformatics Lab Instructions



Drive

Cerca in Drive

lab-gu-pub

Tipo Persone Data modifica Sorgente

Nome

Nome	Proprietario	Data modifica	Dimensione
01-dataset-miRNA.ipynb			
02-neuron-models.ipynb			
03-fmnst-with-population-coding.ipynb			
04-lab.ipynb	gianvito.urgese	11:12	148 kB

This script should be customized for classifying the miRNA expression level dataset.

The task is to implement a classifier of tumour type (optional sex or some other attributes available in the label file).

<https://drive.google.com/drive/folders/1KYh2zT8O-KOAYP7vQ60-oYMzIJu34sbM?dmr=1&ec=wgcdrive-%5Bmodule%5D-goto>

Bioinformatics Lab Instructions

- The full workflow will be used in the Bioinformatics Lab for classification of miRNA expression levels.
- The task is to implement 3 new SNN classifiers of tumour type by exploring:
 - Neuron types
 - Number of layers
 - Number of neuron per layer
- Submit the notebooks (in a zip) with the analysis on the **Elaborati** section of the course



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01URXOV - Machine learning in applications (AA-ZZ) (6 cfu)
Docente: Santa Di Cataldo
Periodo: 2

**Consegna Elaborati**
Tramite questo strumento è possibile consegnare file da parte degli studenti ai docenti del corso in oggetto. Una volta caricato, il file sarà visibile solo al docente ed allo studente che ha eseguito il caricamento e non sarà possibile per lui eliminarlo, ma solo richiedere la cancellazione dei propri elaborati al docente. Tali file resteranno quindi nell'elenco sottostante barrati e contrassegnati dal simbolo ✖, in fase di download verrà inoltre loro aggiunto il prefisso "deleted_".

Neuromorphic Computing & Bioinspired Temporal data

Instruction next week

End